0580_w24_ms_42_Q5

Question	Answer	Marks	Partial Marks	Guidance
5(a)(i)(a)	1480	**1**		
5(a)(i)(b)	440	**2**	**M1** for [UQ =] 1600 soi or [LQ =] 1160 soi	M1 implied by e.g. 1600 – *k* Award M1 for answer 1600 or 1160
5(a)(ii)	$\frac{8}{80}$ oe	**2**	**M1** for 72 or 8 written	Only accept fraction, percentage and decimal equivalents (3sf or better) Ignore incorrect cancelling or conversion following a correct answer. 72 could be on diagram Allow M1 for $\frac{72}{80}$ oe but not for e.g. $\frac{1}{72}$ alone
5(b)	236 or 235.5 to 235.6	**3**	**M2** for $\frac{5.104}{1.3} \times 60$ oe or **M1** for $\frac{5.104}{\textit{their}\text{ time}}$	M2 for e.g. $5.104 \div \frac{78}{3600}$ oe , accept 0.0217 or 0.02166 to 0.02167 for $\frac{78}{3600}$ Condone 5104 used for 5.104 for M1 M1 implied by 3.93 or 3.926… For M1 *their* time could be e.g. 1.18, 78, 1m 18s or these values divided by 60 or (60 × 60) Ignore incorrect attempt at conversion for M1 If using hours allow 0.021 to 0.022 for *their* time
5(c)(i)	(160 – 120) × 0.2 [= 8]	**1**	with no errors seen	Accept 40 × 0.2 nfww Must show multiplication for the mark See AG

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Question	Answer	Marks	Partial Marks	Guidance
5(c)(ii)	22, 36, 64, 30 seen	**B2**	**B1** for 2 or 3 correct frequencies or **M1** for three of 1.1 × (180 – 160), 1.8 × (200 – 180), 1.6 × (240 – 200) and 0.5 × (300 – 240) oe	Allow B2 or B1 if frequency products are used correctly in calculations with *their* midpoints to find Σfm.
	(8 × 140 + *their*22 × 170 + *their*36 × 190 + *their*64 × 220 + *their*30 × 270) ÷ 160	**M3**	**M1** for midpoints soi **M1** for Σfx, x in interval or boundary of interval **M1 dep on second M1** for $\Sigma fx \div 160$	*their* frequencies not fd's and must be integers Midpoints: 140, 170, 190, 220, 270 Condone one error/omission For M1 accept use of expressions such as 120 + (160 – 120)/2 etc for the midpoints. Allow one further error or omission fm: 1120, 3740, 6840, 14080, 8100. Total $\Sigma fm = 33880$
	211.75	**B1**		Condone $200 < v \leq 240$ after 211.75 seen Accept answers 212, 211.7 or 211.8 nfww See AG for alt methods